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Title: Extreme-Aware Loss Functions for LoRA-Adapted Time Series Foundation Models: A Fire Weather Index Case Study

Introduction:

In many time-series forecasting problems, the average day does not matter as much as the rare outliers, and often those outliers are the hardest to predict [1]. Time series foundation models (TSFMs) such as Google's TimesFM and Amazon's Chronos are pretrained on diverse collections of unrelated time series data, and then forecast series they have never seen without being fitted to them [2], [3]. These TSFMs consistently score well on standard aggregated benchmarks for both typical days and outliers [4]. How they perform on outliers specifically is poorly understood, though, with studies across different domains reporting different results [4], [5].

A proposed case study is the Canadian Fire Weather Index (FWI), the most widely used fire danger rating, which recursively accumulates up to nearly two months of dryness, making it a demanding test of whether a TSFM can predict rare extremes on data with long memory [6], [7].

TSFMs' baseline training objectives weight every time interval equally, so non-outlier days dominate training, which pulls their predictions towards typical values and causes them to underpredict rare extremes [1]. For issues like fire danger, those are the ones that matter most, since they are when evacuations are ordered. LoRA, which adapts a pretrained model by training a small set of added parameters, can be implemented, and the loss it is trained with can penalize failures on extreme days more than ordinary days by a fixed rule, by extreme value theory, or by learned weights [1], [8], [9].

To the author's knowledge, no study has investigated which loss characteristics and which data characteristics determine whether LoRA adapted TSFMs capture extremes. This study will test that by predicting FWI across five pyromes using TimesFM and Chronos-2 under different loss functions, and comparing them with a locally trained model.

Research Questions:

This study investigates whether TSFMs adapted using LoRA with extreme-aware loss functions can forecast extreme FWI days, and what determines their success. Extreme-aware loss functions weight errors on extreme days more heavily than errors on ordinary days. Specifically, it addresses the following questions:

1. How do different loss functions (each TSFM's built-in loss, fixed-rule weighting, Extreme Value Loss, and ReFine's learned weighting) affect how well LoRA-adapted TSFMs forecast extreme FWI days, and how much overall accuracy (on ordinary and extreme days) does each one give up?

2. Which site characteristics are associated with how much extreme-aware loss functions improve forecasts of extreme FWI days?
3. How do LoRA-adapted TSFMs trained with extreme-aware loss functions compare with a locally trained LSTM (Long Short Term Memory) in forecasting extreme FWI days?

Literature Review:

TSFMs show a lot of potential in general forecasting accuracy, but results conflict across domains and models. For example, in forecasting wildfire smoke (PM2.5), Huang et al. found that a trained model detected the most hazardous smoke days better than any TSFM [4]. In contrast, Rangaraj et al. found that Chronos beat every other model on forecasting daily water levels in the Everglades wetland, but the other four foundation models did relatively poorly [10]. Meanwhile, Pan and Ezzat found that in forecasting electricity prices, TSFMs were highly competitive with general-purpose baselines [11]. These conflicting results suggest that a variable or set of variables in the data or the training may be determining the success of the model.

One salient aspect of TSFM training is how TSFMs are adapted to a new domain. LoRA is a technique used for this and Gupta et al. showed that it works on data outside a model's pretraining domains with limited data [12]. However, Huang et al. showed that while using LoRA on TSFMs improved them substantially, they still did not beat the trained models, and the authors suggest that this is because LoRA did not teach the models the temporal dynamics that govern extreme events [4]. Notably, their LoRA adapters were trained with each model's baseline loss function, which weights predictions on ordinary and extreme time intervals equally [4].

Loss functions that weight extreme time intervals more heavily than ordinary ones exist, but the efficacy of each has not yet been compared on TSFMs. Ding et al. found that the weakness of deep learning models on extremes comes from minimizing mean squared error, and their Extreme Value Loss technique addresses this issue using Extreme Value Theory [1]. Granted, the quantile loss used by TSFMs such as TimesFM and Chronos-2 penalizes underprediction more for the high-end forecast, but it still weights extreme and ordinary time intervals equally [2], [13]. Shi et al. proposed ReFine, a weighted loss function that gives errors on extreme data samples a bigger penalty than errors on non-extreme data [9]. After creating a reweighted model using meta-learning, ReFine fine-tunes it using only the rare extreme samples, and both these steps work with any neural network [9].

On the data side, some characteristics are already known to affect the performance of TSFMs. Datta Gupta found that TSFMs struggled when their context window missed full cycles of long-period signals [14]. Furthermore, Cohen et al. reported that general TSFMs underperformed on heavy-tailed, nonstationary data [5]. Regarding FWI, what counts as an extreme value differs by climate. For example, an index of 25 in cooler zones would be considered extreme whereas 50 in drier ones would be considered high [6].

Methodology: The method is a quantitative approach.

1. Data Collection

Data from each of the five global pyromes defined by Archibald et al. will be selected using the regions defined in the paper's Figure 2 [16]. Each data location will lie well inside a single pyrome on Archibald et al.'s map, away from boundaries between pyromes. Ten sites per pyrome will be selected by stratified random sampling, using each pyrome as one stratum, with a fixed random seed so the selection is reproducible. Data for each site will come from the ECMWF fire danger reanalysis as it provides both FWI and Drought Code, FWI's component, from 1980 to 2024 [17]. Extreme days for each site will be classified as the days that lie above the 95th percentile in FWI, computed from the training period only and then applied to the validation and test periods, because what is considered extreme in each climate differs [18]. Chronological training will be used for the LoRA-adaptation for the TSFMs. The data from 1980-2011 will be used for training, the data from 2012-2015 will be used for validation, and the years 2016-2024 will be used for testing to roughly follow the 70/10/20 data split defined in time series forecasting literature [19].

2. Models and Baselines

Chronos-2 and TimesFM 2.5 will be adapted using HuggingFace's parameter efficient fine tuning (PEFT) library. Each forecasting configuration will be univariate, meaning it takes in FWI data and FWI data only. The context length will be 1,088 days, which covers roughly three annual cycles and is a multiple of TimesFM's 32-day input patch size, so the models can learn seasonal structures and how structure varies over years. These adapted models will be compared with the zero-shot version of both the TSFMs that were imported, the persistence statistical model, the climatology statistical model, and a locally trained LSTM. A separate LSTM will be trained for each site with each loss function, so every site has its own locally trained model, and the effect of each loss function can be measured per site.

3. LoRA Adaptation and Loss Functions

LoRA adapters will be added to all linear layers in the base TSFMs rather than only the attention layers, to prevent the adaptation ceiling that Huang et al. encountered [4], [8]. A LoRA rank of 8 with an alpha value of 16 will be used across all adapters [4].

Four loss functions will be tested in this study because each one represents a distinct weighting approach: Baseline Loss, Fixed Rule [20], Extreme Value Loss [1], and ReFine [9].

Extreme Value Loss's weighting is applied to each of the quantile loss terms for the TSFMs. For ReFine, the weights are optimized during training against a set of extreme events drawn from the validation period, and the model is then fine-tuned on the extreme days from the training period

only, with both the reweighting and the fine-tuning updating the same LoRA adapter for the TSFMs.

4. Data Characteristics

Four quantitative characteristics are computed for each dataset from the training period, so that the forecasting results can be examined for patterns in the kinds of sites that benefit most from extreme-aware loss functions. The pyrome of the dataset is also recorded. The characteristics recorded are Drought Code accumulation- the mean annual maximum Drought Code at the site; tail heaviness- the 99th percentile divided by the median; seasonality strength- the share of variance in the dataset explained by the annual cycle; and clustering of extreme days- the mean length of periods of two or more consecutive days above the site's 95th percentile threshold, over the training period.

5. Evaluation

Each model will forecast at a 7-day horizon to match the outlooks that agencies produce daily for resource allocation and prepositioning [21], [22]. A day is predicted as extreme when the model's forecast crosses the site's extreme threshold. To measure whether a model detects rare events, it will be scored using SEDI (Symmetric Extremal Dependence Index) which is based on both the hit rate and the false alarm rate [23]. To measure the cost of the loss functions which weight extreme days more heavily, MAE (Mean Absolute Error) will be computed. The TSFMs and the LSTM will be scored on SEDI and MAE under each of the four loss functions, so that the gain on extreme days can be weighed against the cost in overall accuracy. Each site's SEDI gain of an extreme-aware loss function over the baseline loss function will be correlated with each of the four data characteristics across sites using Spearman rank correlation, and the gains will also be compared across pyromes. Lastly, to compare the LoRA-adapted TSFMs with the LSTM and the reference forecasts, each combination of model and site will be scored (SEDI, MAE), with the median taken across sites to summarize the results into a general comparison.

Expected Outcome:

The study will produce a comparison of four loss functions across two TSFMs and a locally trained LSTM at 50 sites spanning five pyromes (10 sites per pyrome). The comparison will show which weighting approach best improves the detection of extreme fire days, and what it costs on overall accuracy (ordinary and extreme days). The comparison will also identify which site characteristics correlate with the largest gains in extreme-day forecasting. If no extreme-aware loss function improves extreme-day detection, the LSTM results will help explain why. If these loss functions improve the LSTM but not the LoRA-adapted TSFMs, that suggests LoRA's capacity is the binding constraint. If they improve neither, that suggests the FWI history alone does not contain enough information to accurately forecast extreme days.

Potential Limitation:
One potential limitation is that every model is being configured with a univariate input. Models see only FWI history, so extremes driven by sudden weather shifts are partly unpredictable. Univariate input is also the only configuration both TSFMs share, since TimesFM is natively a univariate model [2], and it keeps the loss function as the only thing that varies, although skill may be limited for every model at a 7-day horizon. Furthermore, each FWI value is an average of the FWI over a grid cell of approximately 31 km which may make the highest local extreme smooth out because of the averaging. 50 sites is a modest sample meaning that any correlations between the site characteristics and extreme-day prediction should be treated as exploratory rather than absolute. Lastly, this study does not compare the forecasted results to GEFF forecasting [15] which forecasts using full weather models, so no claim in this paper is made about operational feasibility or performance.

Project Practicalities:
This project will span approximately six months. This timeline will span 1) site selection and downloading the FWI and Drought Code series, along with computing each site's extreme-day threshold and characteristics (1 month); 2) building the forecasting pipeline (1 month); 3) adapting the TSFMs and training the LSTMs under each loss function across all sites (2 months); 4) computing MAE and SEDI, running the correlation analysis between the extreme-day detection and site characteristics, and preparing the report (2 months).

Conclusion:
TSFM behavior on predicting rare extremes is poorly understood, and no study has compared loss function approaches on LoRA-adapted TSFMs. This study adapts two TSFMs with LoRA using four different loss functions across five pyromes, each having 10 sites. This study would show which loss function characteristics and qualities of a dataset drive success in forecasting rare extremes. That would tell people in other fields with rare extremes whether this approach is worth trying, without each of them needing to test it from scratch.

Post-program Plan:
Upon completion, the author intends to test whether an adapter trained in some pyromes can transfer to others, and to add weather inputs as covariates to improve forecasting. The author also intends to present the findings at academic research conferences or publish them in relevant journals. Comparison against real-world operational forecasts such as GEFF and extension to other domains with rare extremes such as marine heatwaves are also possible extensions of this work.

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